Program overview

19-May-2025 16:35

Year	**2024/2025**
Organization	**Mechanical Engineering**
Education	**Bachelor Mechanical Engineering**

Code	Omschrijving	ECTS	p1 p2 p3 p4 p5
Bachelor WB 2024			
1e jaar BSc WB 2024	**1st Year BSc WB 2024**		
Wiskunde blok			
WBMT1050-24	Mathematics 1	6	
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WBMT1050-24 T3	*Analysis-preknowledge test*	*0*	
WBMT1051	Mathematics 2	6	
WBMT1051 Toets 1	*Lineaire Algebra 1 - deeltentamen*	*3*	
WBMT1051 Toets 2	*Lineaire Algebra 2 - deeltentamen*	*3*	
Werktuigbouwkunde Theorie blok			
WB1135	Introductory Dynamics	6	
WB1530-14	Thermodynamics	6	
WB1630-20	Statics	6	
WB1631-24	Mechanics of Materials	6	
Werktuigbouwkunde Project blok			
WB1641	Design Engineering Project 1	6	
WB1641 Toets 1	*Projecttentamen*	*3*	
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WB1642 Toets 1	*Projecttentamen*	*3*	
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WB1643A	Design Engineering Project 3A	6	
WB1643A Toets 1	*Projecttentamen*	*3*	
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2e jaar BSc WB 2024	**2nd Year BSc WB 2024**		
WB201-11 Wiskunde blok	**Mathematics block**		
WBMT2048	Mathematics 3	6	
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Werktuigbouwkunde Theorie blok			
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WB2630	Advanced Mechanics	6	
WB2630 Toets 1	*Rigid-Body Dynamics*	*3*	
WB2630 Toets 2	*Continuum Mechanics*	*3*	
Werktuigbouwkunde Project blok			
WB2232	Project Robotics	6	
WB2232 Toets 1	*Robotics exam*	*3*	
WB2232 Toets 2	*Groepswerk*	*3*	
WB2332	Project Materials Sciences	6	
WB2332 Toets 1	*Materiaalkunde practicum + projecttoets*	*3*	
WB2332 Toets 2	*Groepswerk*	*3*	
WB2543	Process Engineering & Thermodynamics	6	
WB2543 Toets 1	*PE&T tentamen*	*3*	
WB2543 Toets 2	*Groepswerk + practicum*	*3*	
WB2633	Advanced Engineering Design Project	6	
WB2633 T1	*AED Exam*	*2*	
WB2633 T2	*FEM-practical*	*1*	
WB2633 T3	*AED Project*	*3*	
3e jaar BSc WB 2024	**3rd Year BSc WB 2024**		

3e jaar WB minors 2024	**3rd Year WB minors 2024**	
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WB3135	Integrated Mechanical Systems	6
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WB3135 Toets 2	*IMS - tentamen*	*3*
WB3240-24	Systems and Control Engineering	6
WB3240-24 T1	*Systems and Control Engineering*	*2*
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Eindproject blok		
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WB3073 Toets 1	*I&S Exam*	*2*
WB3073 Toets 2	*Group Essay*	*2*
WB3BEP-16	Bachelor End Project	14

Year	**2024/2025**
Organization	**Mechanical Engineering**
Education	**Bachelor Mechanical Engineering**

Bachelor WB 2024	
Director of Studies	Dr. R. Delfos
Program Title	Bachelor Mechanical Engineering
Director of Education	Director of Studies Bachelor Mechanical Engineering: Dr. René Delfos
Program Coordinator	onderwijsorganisatie-ME@tudelft.nl
Introduction 1	Please switch to Dutch language

Year	**2024/2025**
Organization	**Mechanical Engineering**
Education	**Bachelor Mechanical Engineering**

1e jaar BSc WB 2024

Year	**2024/2025**
Organization	**Mechanical Engineering**
Education	**Bachelor Mechanical Engineering**

Wiskunde blok

WBMT1050-24	**Mathematics 1**	**6**
Responsible Instructor	Dr.ir. D. den Ouden-van der Horst	
Instructor	Dr.ir. N.P.K. Chan	
Contact Hours / Week x/x/x/x	4/4/X/X	
Education Period	1 2	
Start Education	1	
Exam Period	1 2 3	
Course Language	Dutch	

WBMT1050-24 T1	***Analysis 1***	***3***
Responsible Instructor	Dr.ir. D. den Ouden-van der Horst	
Instructor	Dr.ir. N.P.K. Chan	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	Dutch	

WBMT1050-24 T2	***Analysis 2***	***3***
Responsible Instructor	Dr.ir. D. den Ouden-van der Horst	
Instructor	Dr.ir. N.P.K. Chan	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	Dutch	

WBMT1050-24 T3	***Analysis-preknowledge test***	***0***
Responsible Instructor	Dr.ir. D. den Ouden-van der Horst	
Instructor	Dr.ir. N.P.K. Chan	
Education Period	1 2	
Start Education	1	
Exam Period	1 1A 2A	
Course Language	Dutch	

WBMT1051	**Mathematics 2**	**6**
Responsible Instructor	Dr.ir. D. den Ouden-van der Horst	
Co-responsible for assignments	Dr. O. Jaibi	
Education Period	3 4	
Start Education	3 4	
Exam Period	3 4 5	
Course Language	Dutch	

WBMT1051 Toets 1	***Lineaire Algebra 1 - deeltentamen***	***3***
Responsible Instructor	Dr.ir. D. den Ouden-van der Horst	
Instructor	Dr. O. Jaibi	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	Dutch	

WBMT1051 Toets 2	*Lineaire Algebra 2 - deeltentamen*	*3*
Responsible Instructor	Dr.ir. D. den Ouden-van der Horst	
Instructor	Dr. O. Jaibi	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	Dutch	

Werktuigbouwkunde Theorie blok

WB1135	Introductory Dynamics	6
Responsible Instructor	Prof.dr. P.G. Steeneken	
Instructor	F.G.J. Broeren	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	Dutch	
Department	ME Department Precision & Microsystems Engineering	

WB1530-14	Thermodynamics	6
Responsible Instructor	Prof.dr. R. Pecnik	
Instructor	Dr.ir. W.P. Breugem	
Instructor	N.F. Herting	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	Dutch	
Department	ME Department Process & Energy	

WB1630-20	Statics	6
Responsible Instructor	Prof.dr.ir. J.L. Herder	
Contact Hours / Week x/x/x/x	8/X/X/X	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	Dutch	
Department	ME Department Precision & Microsystems Engineering	

WB1631-24	Mechanics of Materials	6
Responsible Instructor	Prof.dr.ir. M. Langelaar	
Contact Hours / Week x/x/x/x	X/8/X/X	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	Dutch	
Course Contents	See Dutch version.	
Study Goals	See Dutch version.	
Education Method	See Dutch version.	
Assessment	See Dutch version.	
Department	ME Department Precision & Microsystems Engineering	

Year	**2024/2025**
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Werktuigbouwkunde Project blok

WB1641	Design Engineering Project 1	6
Responsible Instructor	Dr.ir. R.W. Vroom	
Responsible Instructor	Dr.ir. G. Radaelli	
Education Period	1	
Start Education	1	
Exam Period	Different, to be announced	
Course Language	Dutch	
Department	ME Department Precision & Microsystems Engineering	

WB1641 Toets 1	*Projecttentamen*	*3*
Responsible Instructor	Dr.ir. R.W. Vroom	
Responsible Instructor	Dr.ir. G. Radaelli	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	Dutch	
Department	ME Department Precision & Microsystems Engineering	

WB1641 Toets 2	*Groepswerk en individuele vaardigheden*	*3*
Responsible Instructor	Dr.ir. R.W. Vroom	
Responsible Instructor	Dr.ir. G. Radaelli	
Education Period	1	
Start Education	1	
Exam Period	none	
Course Language	Dutch	
Department	ME Department Precision & Microsystems Engineering	

WB1642	Design Engineering Project 2	6
Responsible Instructor	Dr.ir. R.W. Vroom	
Education Period	2	
Start Education	2	
Exam Period	Different, to be announced	
Course Language	Dutch	
Department	ME Department Precision & Microsystems Engineering	

WB1642 Toets 1	*Projecttentamen*	*3*
Responsible Instructor	Dr.ir. R.W. Vroom	
Responsible Instructor	Dr.ir. G. Radaelli	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	Dutch	
Department	ME Department Precision & Microsystems Engineering	

WB1642 Toets 2	*Groepswerk en individuele vaardigheden*	*3*
Responsible Instructor	Dr.ir. R.W. Vroom	
Responsible Instructor	Dr.ir. G. Radaelli	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	Dutch	
Department	ME Department Precision & Microsystems Engineering	

WB1643A	**Design Engineering Project 3A**	**6**
Responsible Instructor	Dr.ir. R.W. Vroom	
Responsible Instructor	Dr.ir. G. Radaelli	
Education Period	3	
Start Education	3	
Exam Period	Different, to be announced	
Course Language	Dutch	
Department	ME Department Precision & Microsystems Engineering	

WB1643A Toets 1	***Projecttentamen***	***3***
Responsible Instructor	Dr.ir. R.W. Vroom	
Responsible Instructor	Dr.ir. G. Radaelli	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	Dutch	
Department	ME Department Precision & Microsystems Engineering	

WB1643A Toets 2	***Groepswerk en individuele vaardigheden***	***3***
Responsible Instructor	Dr.ir. R.W. Vroom	
Responsible Instructor	Dr.ir. G. Radaelli	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	Dutch	
Department	ME Department Precision & Microsystems Engineering	

WB1643B	**Design Engineering Project 3B**	**6**
Responsible Instructor	Dr.ir. R.W. Vroom	
Responsible Instructor	Dr.ir. G. Radaelli	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	Dutch	
Department	ME Department Precision & Microsystems Engineering	

WB1643B Toets 1	***Projecttentamen***	***3***
Responsible Instructor	Dr.ir. R.W. Vroom	
Responsible Instructor	Dr.ir. G. Radaelli	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	Dutch	
Department	ME Department Precision & Microsystems Engineering	

WB1643B Toets 2	***Groepswerk en individuele vaardigheden***	***3***
Responsible Instructor	Dr.ir. R.W. Vroom	
Responsible Instructor	Dr.ir. G. Radaelli	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	Dutch	
Department	ME Department Precision & Microsystems Engineering	

Year	**2024/2025**
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2e jaar BSc WB 2024	
Prerequisites	To be able to participate in this course you must register in time. Registration for a course is not the same as registration for a (partial) exam, registration in bright space or registration for a study link. More information about registering for courses, the procedure and current deadlines can be found at; https://www.tudelft.nl/index.php?id=50357&L=1. Registration for exams must be done separately. * in some cases there are entry requirements or additional criteria to be admitted to a course. These can be found in the TER, https://www.tudelft.nl/index.php?id=33610&L=1.

Year	**2024/2025**
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WB201-11 Wiskunde blok

WBMT2048	Mathematics 3	6
Responsible Instructor	Dr.ir. R.F. Swarttouw	
Education Period	1 2	
Start Education	1 2	
Exam Period	Different, to be announced	
Course Language	Dutch	
Required for	(Fluid) Mechanics, Mechatronics.	
Expected prior knowledge	First year Mathematics courses.	
Course Contents	WBMT2048 T1 - Analysis 3: - Series and sequences; - Line and surface integrals; - Integral Theorems of Green, Stokes, and Gauss. WBMT2048 T2 - Differential Equations: - Linear first and second order equations; - Laplace transform; - Systems of linear equations - Autonomous systems and stability; - Fourier series; - Partial differential equations.	
Study Goals	Study goals for each part can be found on the corresponding StudyGuide-page (WBMT2048 Toets 1 or WBMT2048 Toets 2).	
Education Method	Lectures 4 hours per week.	
Literature and Study Materials	WBMT2048 Toets 1: Calculus, Volume 1, 2 and 3. Edwin Herman, Gilbert Strang. ISBN-13 978-1-947172-13-5, 978-1-947172-14-2, 978-1-947172-16-6 This is an open source book that will also be made available via Brightspace. WBMT2048 Toets T2: William E. Boyce, Richard C. DiPrima & Douglas B. Meade: Elementary Differential Equations and Boundary Value Problems. International Adaption, Twelfth Edition, Wiley, 2022. ISBN 978-1-119-82051-2.	
Assessment	WBMT2048 T1: - Written exams (see Brightspace for more details). - There will be two partial exams: one partial exam in week 5 covering the first part of the course (Vector Calculus) and one partial exam in week 10 covering the second part of the course (Sequences and Series). Both exams last 90 minutes. The final grade for this course is the average of both grades for the partial exams. There is one resit, covering the entire course, which lasts 180 minutes. It is not possible to resit only one partial exam. WBMT2048 T2: - Written exams (see Brightspace for more details).	
Exam Hours	WBMT2048 T1: 2 partial exams of 90 minutes each. One resit of 180 minutes. WBMT2048 T2: 180 minutes.	
Permitted Materials during Tests	WBMT2048 T1: A formula sheet (see Brightspace). NO calculators. WBMT2048 T2: A table of Laplace transforms (see Brightspace). NO calculators.	
Enrolment / Application	See Dutch comment.	
Judgement	See Dutch comment.	
Contact	WBMT2048@tudelft.nl	

WBMT2048 Toets 1	Analyse - deeltentamen	3
Responsible Instructor	Dr.ir. R.F. Swarttouw	
Instructor	Drs. A.T. Hensbergen	
Education Period	1	
Start Education	1	
Exam Period	2 1A 1B	
Course Language	Dutch	
Required for	(Fluid) Mechanics, Mechatronics.	
Expected prior knowledge	First year Mathematics courses: Analysis 1 and 2.	
Course Contents	- Line and surface integrals; - Integral Theorems of Green, Stokes, and Gauss; - Series and sequences.	
Study Goals	After successfully finishing this course, you are able to: - find parametrisations of curves and surfaces (in 2D and 3D); - calculate a line integral or a surface integral of a function or a vector field; - mention the physical interpretation of these line and surface integrals; - define and formulate the physical meaning of concepts like gradient, curl, and divergence; - calculate line and surface integrals via one of the three integral theorems (Green, Stokes, and Gauss); - determine whether a sequence is convergent or divergent, for example by calculating its limit; - determine whether a series is convergent or divergent with a test for convergence: Integral Test, (Limit) Comparison Test or Ratio Test; - determine the interval of convergence of a power series; - determine the Taylor series of a simple analytic function, for example sin(x), exp(x), ln(x), arctan(x), (1+x)^a.	
Education Method	2 x 2 hour lectures per week.	
Books	Calculus, Volume 1, 2 and 3. Edwin Herman, Gilbert Strang. ISBN-13 978-1-947172-13-5, 978-1-947172-14-2, 978-1-947172-16-6 This is an open source book that also will be made available (for free) via Brightspace.	
Assessment	- Written exams (see Brightspace for more details). - There will be two partial exams: one partial exam in week 5 covering the first part of the course (Vector Calculus) and one partial exam in week 10 covering the second part of the course (Sequences and Series). - Both exams last 90 minutes. The final grade for this course is the average of both grades for the partial exams. There is one resit, covering the complete course, which lasts 180 minutes. It is not possible to resit only one partial exam.	
Exam Hours	Two partial exams of 90 minutes each. One resit of 180 minutes.	
Permitted Materials during Tests	A formula sheet (see Brightspace). NO calculators.	
Enrolment / Application	See Dutch comment.	
Judgement	This is half of the course WBMT2048. Partial grades for WBMT2048/T1 (Analysis 3) and WBMT2048/T2 (Differential Equations) will be given in one decimal. The final grade for WBMT2048 is the average of the partial grades for WBMT2048/T1 and WBMT2048/T2. The lowest grade needed for averaging is 5,0. Example: 5,0 for T1 + 7,8 for T2 = 12,8 (Total). So this gives an average of 6,4, your final grade for WBMT2048. The lowest Total for a valid Final Grade (a 'pass') is 11,5 (average 5,75). The best result ever achieved in MyTUDelft/Osiris for a partial grade counts.	
Contact	WBMT2048@tudelft.nl	

WBMT2048 Toets 2	***Differentiaalvergelijkingen - deeltentamen***	***3***
Responsible Instructor	Dr.ir. R.F. Swarttouw	
Instructor	Dr. O. Jaibi	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	Dutch	
Expected prior knowledge	First year Mathematics courses: Analysis 1 and Linear Algebra 1 and 2.	
Course Contents	- First-order differential equations (linear, separable, autonomous, direction fields); - Second-order linear differential equations (Wronskian, method of reduction of order, Euler equation); - Laplace transforms and initial value problems (step function, impulse functions, the convolution integral); - Systems of first-order linear differential equations; - Non-linear autonomous systems (locally linear systems, stability, phase plane); - Partial differential equations (Heat equation, Wave equation, Laplace's equation); - Method of separation of variables; - Boundary value problems; - Fourier series (even and odd functions).	
Study Goals	After successfully finishing this course, you are able to: - distinguish between ordinary and partial differential equations; - distinguish between linear and nonlinear differential equations; - solve linear and separable first-order differential equations; - solve autonomous first-order differential equations and determine the behavior near equilibrium solutions; - solve homogeneous linear differential equations with constant coefficients; - solve the Euler equation; - determine particular solutions of non-homogeneous linear differential equations with the method of undetermined coefficients; - use Laplace transforms to solve initial value problems of linear differential equations with constant coefficients; - solve systems of homogeneous linear first-order differential equations with constant coefficients; - determine particular solutions of systems of non-homogeneous linear first-order differential equations with the method of undetermined coefficients; - determine particular solutions of systems of non-homogeneous linear first-order differential equations with the method of variation of parameters; - determine the behavior of solutions near equilibrium points of autonomous systems of differential equations; - determine the Fourier (cosine- and/or sine-)series of a function; - solve the heat or diffusion equation with the method of separations of variables; - solve the wave equation with the method of separations of variables; - solve the Laplace equation with the method of separations of variables.	
Education Method	2 x 2 hour lectures per week.	
Books	William E. Boyce, Richard C. DiPrima & Douglas B. Meade: Elementary Differential Equations and Boundary Value Problems. International Adaption, Twelfth Edition, Wiley, 2022. ISBN 978-1-119-82051-2.	
Assessment	Written exams (see Brightspace for more details). The exam lasts 180 minutes.	
Permitted Materials during Tests	A table with Laplace transforms (see Brightspace). Bring your own printed version to the exam!	
Judgement	This is half of the course WBMT2048. Partial grades for WBMT2048/T1 (Analysis 3) and WBMT2048/T2 (Differential Equations) will be given in one decimal. The final grade for WBMT2048 is the average of the partial grades for WBMT2048/T1 and WBMT2048/T2. The lowest grade needed for averaging is 5,0. Example: 5,0 for T1 + 7,8 for T2 = 12,8 (Total). So this gives an average of 6,4, your final grade for WBMT2048. The lowest Total for a valid Final Grade (a 'pass') is 11,5 (average 5,75). The best result ever achieved in MyTUDelft/Osiris for a partial grade counts.	
Contact	WBMT2048@tudelft.nl	

WBMT2049	**Mathematics 4**	**6**
Instructor	Dr. N.V. Budko	
Instructor	Dr. C.O. Smet	
Education Period	3 4	
Start Education	3 4	
Exam Period	3 4 5 Different, to be announced	
Course Language	Dutch	

WBMT2049 T1	*Kansrekening en Statistiek - deeltentamen*	*3*
Responsible Instructor	Dr.ir. R. Versendaal	
Instructor	Dr. C.O. Smet	
Education Period	3	
Start Education	3	
Exam Period	4 3A 3B	
Course Language	Dutch	

WBMT2049 T2	*Numerieke Wiskunde - deeltentamen*	*3*
Responsible Instructor	Dr. N.V. Budko	
Instructor	D. Toshniwal	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	Dutch	

WBMT2049 T3	*Numerieke Wiskunde - practicum*	*0*
Responsible Instructor	Dr.ir. Z.J.G. Gromotka	
Instructor	D. Toshniwal	
Education Period	4	
Start Education	4	
Exam Period	Different, to be announced	
Course Language	Dutch	

Year	**2024/2025**
Organization	**Mechanical Engineering**
Education	**Bachelor Mechanical Engineering**

Werktuigbouwkunde Theorie blok

WB2235	Signal Analysis	6
Responsible Instructor	Dr.ing. R. van de Plas	
Instructor	G. de Albuquerque Gleizer	
Instructor	Dr. N.J. Myers	
Contact Hours / Week x/x/x/x	0/0/8/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	WI2030WBMT Wiskunde 3 (Differentiaalvergelijkingen) WI2031WBMT Wiskunde 4 (Kansrekening en Statistiek) should be heard in parallel to the course	
Course Contents	This course treats the analysis, filtering and detection of signals influenced by linear time-invariant systems as they occur in typical mechanical and mechatronic applications. Signals will be considered in both deterministic and stochastic formulations. Special emphasis is put on discrete-time techniques that can be easily implemented in practice. The main topics of the course are: 1. Elementary properties of signals and systems (even, odd, linear, stable, causal, linear-phase, group-delay, all-pass, minimum-phase) 2. Fourier and Hilbert transform, one and two-sided Laplace and Z-transforms 3. Bode plots and Bodes phase-gain relationship 4. Continuous-time low-pass filters 5. Modulation and demodulation 6. Analog-to-digital and digital-to-analog conversion 7. Design of discrete-time filters using windowing and impulse invariance 8. Elementary properties of random processes (i.i.d., w.s.s., joint w.s.s., independence); key indicators such as mean, auto and cross-correlation/covariance, and spectral densities 9. Identification of linear time-invariant discrete-time systems in the frequency domain 10. Discrete-time Wiener filtering (finite impulse response and non-causal) 11. Detection of discrete-time signals in Gaussian noise	
Study Goals	Students can 1. Evaluate if a signal or system possesses an elementary property; utilize these in simple problems; give examples that exhibit desired elementary properties 2. Compute Fourier, Laplace, z and Hilbert transforms, both directly and using transform pairs and properties 3. Sketch Bode plots for rational transfer functions 4. Determine the specifications of continuous-time low-pass filters from their frequency response; design low-pass filters that fulfill given specifications 5. Analyze typical modulation and demodulation schemes; add missing components 6. Evaluate the impact of A/D and D/A conversion schemes; choose sampling intervals 7. Design discrete-time filters using impulse invariance, windowing, and Wiener techniques 8. Evaluate if a random process possesses an elementary property; utilize them in simple problems; determine key indicators of random processes 9. Identify the frequency response of a linear time-invariant discrete-time system from actual measurements; specify the bias and the variance 10. Derive optimal detection rules for a known signal in Gaussian noise	
Education Method	Lectures, problem sets and practice sessions (werkcolleges).	
Literature and Study Materials	- A.V. Oppenheim, A.S. Willsky and S. Hamid, Signals and Systems, Pearson New International Edition, 2nd ed, 2013. - A.V. Oppenheim and G. Verghese, Signals, Systems and Inference, Class Notes for 6.011: Introduction to Communication, Control and Signal Processing, MIT, Spring 2010. Available online: http://ocw.mit.edu/courses/electrical-engineering-and-computer-science/6-011-introduction-to-communication-control-and-signal-processing-spring-2010/readings/ - For some parts of the lecture, external materials will be used. These materials will either be available online from within the network of TU Delft, or be provided via Brightspace.	
Assessment	Written exam. Students will only be allowed to bring a non-programmable and non-graphing calculator. A formula sheet will be provided.	
Department	ME Department Delft Center for Systems and Control	

WB2330	Materials Sciences	6
Responsible Instructor	Dr.ir. S.E. Offerman	
Instructor	Dr. G.A. Filonenko	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	Dutch	
Department	ME Department Materials Science & Engineering	

WB2542	Fluid Mechanics & Heat Transfer	6
Responsible Instructor	Dr.ir. G.E. Elsinga	
Education Period	1 2	
Start Education	1	
Course Language	Dutch	
Department	ME Department Process & Energy	

WB2542 Toets 1	*Stromingsleer*	*3*
Responsible Instructor	Dr. D.S.W. Tam	
Instructor	Prof.dr.ir. C. Poelma	
Education Period	1	
Start Education	1	
Exam Period	2 1A 1B	
Course Language	Dutch	
Department	ME Department Process & Energy	

WB2542 Toets 2	*Warmte-overdracht*	*3*
Responsible Instructor	Dr.ir. G.E. Elsinga	
Instructor	Dr.ir. M.J. Tummers	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	Dutch	
Department	ME Department Process & Energy	

WB2630	**Advanced Mechanics**	**6**
Responsible Instructor	Dr.ir. L.F.P. Noel	
Responsible Instructor	Dr.ir. A.H.A. Stienen	
Education Period	1 2	
Start Education	1	
Exam Period	1 2 3	
Course Language	Dutch	
Course Contents	See the respective parts T1 and T2	
Study Goals	See the respective parts T1 and T2	
Education Method	See the respective parts T1 and T2	
Assessment	See details in T1 and T2	
Enrolment / Application	see Dutch version	
Department	ME Department Biomechanical Engineering ME Department Precision & Microsystems Engineering	
Judgement	see Dutch version	

WB2630 Toets 1	***Rigid-Body Dynamics***	***3***
Responsible Instructor	Dr.ir. A.H.A. Stienen	
Instructor	Ir. P.H. de Jong	
Contact Hours / Week x/x/x/x	lectures 2/0/0/0 Practicals 2/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	Dutch	
Expected prior knowledge	Linear Algebra, Dynamics (2D)	
Course Contents	1) Kinematics of a rigid body in 3D a. Description of single and successive 3D rotations via Euler angles, rotation matrices, axis and angle of rotation. b. Angular velocity and instantaneous axis of rotation c. Transport Theorem 2) Kinetics of a rigid body in 3D a. The inertia tensor b. Principle of work & energy c. Newton-Euler equations d. Free rotational motion, gyroscopic effects e. Numerical integration 3) Systems with n degrees of freedom a. Generalized coordinates b. Principle of virtual work c. Lagranges equations 4) Methods to check plausibility of generated results, such as a. Units notation and checks b. Numerical approximation c. Checking validity of assumptions d. Treatment of special cases	
Study Goals	Kinematics of a rigid body in 3D The student is able to describe rotations of a rigid body in 3D via Euler angles, rotation matrices, and the principal axis of rotation. transform one description of rotations into the other, and re-write successive rotations as a single rotation in the different representations. describe angular velocity of a rigid body in 3D and find the instantaneous axis of rotation apply transport theorem to take derivatives of vectors in rotating frames Kinetics of a rigid body in 3D The student is able to calculate the inertia tensor for a rigid body via integration analyze the inertia tensor and find the principal axes derive the equations of motion for a rigid body in 3D using the principle of work and energy or using the Newton-Euler equations recognize and explain gyroscopic effects Systems with n degrees of freedom The student is able to formulate kinetic and potential energy for systems of particles and for rigid bodies in generalized coordinates apply Lagranges methods to simple systems of particles and to rigid bodies to obtain the equations of motion recognize conservative systems Generating reliable results to mechanical problems The student is able to test the plausibility of calculation steps and final results be aware of and check validity of assumptions whenever applying learned concepts	
Education Method	Each week, interactive on-campus lectures and tutored sessions are organized. Students study the relevant course material before each lecture. For this, they can use the information and material provided on Brightspace and the recommended books. During the tutored sessions, students can work on homework exercises and ask for further explanation.	
Books	Vallery/Schwab "Advanced Dynamics", in 1st (black), 2nd (blue), or 3rd (red) edition	
Assessment	The assessment for this course includes a digital exam in week 9 (30%) and a written exam in week 10 (70%). Both exams are conducted on-campus and must be taken individually. There is no minimum required grade for either exam. The final grade is calculated as the weighted average of both exams. A 5.0 or higher is needed for the final grade for WB2630 T1 before it can be combined in the WB2630 (T1+T2) result. The resit for the course combines both exams into a single sitting in the following quarter. The exams are organized on-campus, are proctored, and are closed-book. Students are allowed to bring a reference sheet (one A4 page, double-sided, handwritten by the student themselves, not copied or resized). For information on how the grades for WB2630-T1 (Rigid Body Dynamics) and WB2630-T2 (Continuum Mechanics) are combined to form the final grade for WB2630 (Advanced Mechanics), please refer to the Brightspace page for WB2630.	
Enrolment / Application	See Dutch version	
Department	ME Department Biomechanical Engineering	
Judgement	See Dutch version	
Contact	Rigid-Body-Dynamics-ME@tudelft.nl. Please do not use lecturers' individual e-mail addresses.	

WB2630 Toets 2	Continuum Mechanics	3

Responsible Instructor Dr.ir. L.F.P. Noel

Contact Hours / Week x/x/x/x lectures 0/2/0/0 Practicals 0/2/0/0 Office houres 0/2/0/0

Education Period 2

Start Education 2

Exam Period 2
3

Course Language English

Course Contents

Linear 3D continuum theory
1. Kinematics equations in 3D
2. Deformation tensor in 3D, linear and nonlinear
3. Stretch in arbitrary direction
4. Equilibrium equations in 3D
5. Stress vector and tensor in 3D, linear
6. Principal stresses and directions, normal and shear stress acting on a plane
7. Constitutive equations in 3D
8. Generalized Hookes law
9. Plane stress/strain state

Energy principles
1. Principle of virtual work (PVW) without/with deformations
2. Principle of minimum potential energy (MPE)
3. Approximate solutions based on MPE

Discrete linear systems
1. Systematic way of solving problems with n degrees of freedom (DOF)
2. Analysis of discrete linear systems (spring, truss) DOF, generalized deformations, generalized stresses, generalized constitutive equations, discrete continuity and equilibrium, member/global stiffness, assembly, boundary conditions, solution

Finite Element Method
1. Approximations based on MPE or PVW
2. Derivation of simple elements and isoparametric elements
3. Analysis of structures with the FEM DOF, element/global stiffness, assembly, boundary conditions, distributed loads, solution
4. Discussion on element choice, mesh refinement, convergence

Study Goals

By the end of the course, the student is able to:

Linear 3D continuum theory
1. Recognize situations where the continuum theory is applicable
2. Derive kinematics equations in 3D
   a. Derive relation between displacements and deformations
   b. Provide interpretation for the components of the strain tensor
   c. Recognize situations where kinematics equations are applicable
3. Use kinematics equations
   a. Determine deformations, a strain tensor for known displacements
   b. Determine deformations, stretches in specified directions
   c. Explain the relevance of and the procedure to measure deformations in specified directions
4. Derive the equilibrium equations in 3D
   a. Explain equilibrium equations in 3D
   b. Explain the relations between external forces, internal forces, and stresses
   c. Recognize situations where equilibrium equations are applicable
5. Use equilibrium equations
   a. Determine a stress vector, a stress tensor
   b. Determine a normal stress, a shear stress
   c. Determine principal stresses and directions
6. Derive generalized Hookes law in 3D
   a. Describe its relevance and limitations
7. Apply generalized Hookes law
8. Identify plane stress or plane strain states

Energy principles
1. Describe the principle of virtual work (PVW)
2. Derive equilibrium equations using the PVW
3. Recognize situations where the PVW is applicable
4. Describe the principle of minimum potential energy (MPE)
5. Derive equilibrium equations using the MPE
6. Recognize situations where the MPE is applicable
7. Build approximate solutions using the MPE

Discrete linear systems
1. Build discrete linear systems for given structures
2. Analyze structures with discrete linear systems
   a. Build local member stiffness matrices
   b. Build global system stiffness matrices
   c. Set up system of equations for discrete linear systems
3. Solve problems using discrete linear systems
   a. Impose boundary conditions
   b. Solve for displacements at joints
   c. Post-process for reaction forces, stresses, and strains

Finite Element Method
1. Describe the basic concepts of the Finite Element Method (FEM)
2. Derive simple and isoparametric elements based on the PVW or on the MPE
   a. Build a geometry and a displacement approximation over an element
   b. Build the stiffness matrix for an element
   c. Build energy conjugated loads for an element
   d. Perform numerical integration over an element
3. Analyze structures with the FEM
   a. Build local element stiffness matrices
   b. Build global structure stiffness matrices

	c. Set up systems of equations for FEM model 4. Solve problems with the FEM a. Compute energy equivalent nodal forces b. Impose boundary conditions c. Solve for displacements at nodes d. Post-process for reaction forces, stresses, and strains 5. Identify appropriate element choice, mesh density, mesh refinement, convergence requirements
Education Method	Contact hours Plenary sessions, including lectures and formative in-class tests (4 hours per week) Questions hours in groups, including practice and exercises (2 hours per weeks) Formative in-class tests Formative in-class tests are organized every week right after the lectures and may lead to a bonus on the final course grade (maximum +1.0 point on the final grade). Bonus credits are only valid for the exam, not for the resit. During the in-class tests, you may collaborate with your fellow students and use the lecture slides. The use of any type of electronic communication (smart phone, texting app, email, social media, etc.) or calculator is considered fraud and will be treated as such. Grade calculation The final grade G is computed based on the exam grade GE and the bonus B(GT) obtained from the test grade GT or based on the resit grade GR only. Provided the exam grade GE is 5.00 or higher, G = GE + B(GT). Otherwise, G = GE. For the resit, the final grade is solely based on the resit grade GR and G = GR. The test grade GT can lead to a bonus B(GT) on the final grade and is based on the average of the best 75% of individual test grades. Missing a test leads to a grade of 0.0 for that particular test. Study materials Lecture slides, exercises instructions, previous exams provided on Brightspace
Assessment	Written exam. Taking part in formative in-class tests can lead to bonus on the final grade.
Department	ME Department Precision & Microsystems Engineering
Contact	continuum-mechanics-ME@tudelft.nl

Year	**2024/2025**
Organization	**Mechanical Engineering**
Education	**Bachelor Mechanical Engineering**

Werktuigbouwkunde Project blok

WB2232	Project Robotics	6
Responsible Instructor	Prof.dr.ir. M. Wisse	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	Dutch	
Course Contents	Please consult the course information in Dutch.	
Study Goals	Please consult the course information in Dutch.	
Education Method	Please consult the course information in Dutch.	
Books	"Electronics - a systems approach" (6th edition), Neil Storey	
Assessment	Please consult the course information in Dutch.	
Enrolment / Application	See Dutch version for rules of enrolment.	
Remarks	The course runs tightly integrated with the Signal Analysis course, thus students are expected to follow both courses simultaneously.	
Department	ME Department Cognitive Robotics	
Judgement	See https://www.tudelft.nl/studenten/me-studentenportal/onderwijs/gerelateerd/regelingen-reglementen	

WB2232 Toets 1	*Robotics exam*	*3*
Responsible Instructor	Prof.dr.ir. M. Wisse	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	Dutch	
Course Contents	See general description for detailed info.	
Study Goals	See general description for detailed info.	
Education Method	See the general (Dutch) course description.	
Assessment	See the general (Dutch) course description.	
Enrolment / Application	See Dutch version	
Remarks	See the general (Dutch) course description.	
Department	ME Department Cognitive Robotics	
Judgement	See Dutch version	

WB2232 Toets 2	*Groepswerk*	*3*
Responsible Instructor	Prof.dr.ir. M. Wisse	
Instructor	Ir. C.C. Claij	
Education Period	3	
Start Education	3	
Exam Period	Different, to be announced	
Course Language	Dutch	
Course Contents	See course description for details.	
Study Goals	See course description for details.	
Education Method	Project	
Assessment	See (Dutch) general course description.	
Enrolment / Application	See Dutch version.	
Remarks	See (Dutch) general course description.	
Department	ME Department Cognitive Robotics	
Judgement	See Dutch version	

WB2332	**Project Materials Sciences**	**6**
Responsible Instructor	Y. Gonzalez Garcia	
Instructor	K.R. Rossi	
Instructor	Ir. T.B. Keesom	
Contact Hours / Week x/x/x/x	0/0/0/x	
Education Period	4	
Start Education	4	
Exam Period	Different, to be announced	
Course Language	Dutch English	
Course Contents	See Dutch description	
Study Goals	See Dutch description	
Education Method	See Dutch description	
Assessment	See Dutch description	
Enrolment / Application	See Dutch description	
Department	ME Department Materials Science & Engineering	
Judgement	See Dutch description	

WB2332 Toets 1	***Materiaalkunde practicum + projecttoets***	***3***
Responsible Instructor	Y. Gonzalez Garcia	
Instructor	K.R. Rossi	
Instructor	Ir. T.B. Keesom	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	Dutch	
Course Contents	See Dutch description	
Study Goals	See Dutch description	
Department	ME Department Materials Science & Engineering	

WB2332 Toets 2	***Groepswerk***	***3***
Responsible Instructor	Y. Gonzalez Garcia	
Instructor	K.R. Rossi	
Instructor	Ir. T.B. Keesom	
Education Period	4	
Start Education	4	
Exam Period	Different, to be announced	
Course Language	Dutch	
Course Contents	See Dutch description	
Study Goals	See Dutch description	
Department	ME Department Materials Science & Engineering	

WB2543	Process Engineering & Thermodynamics	6
Responsible Instructor	Prof.dr.ir. J.T. Padding	
Instructor	Dr. R. Delfos	
Instructor	Prof.dr.ir. T.J.H. Vlugt	
Responsible for assignments	L. Botto	
Co-responsible for assignments	Dr.ir. T.M.J. Nijssen	
Education Period	2	
Start Education	2	
Exam Period	Different, to be announced	
Course Language	Dutch	
Tags	Design Energy Energy & Industry Fluid Mechanics Group work Modelling Practicals Process Project Small groups Sustainability Technology Thermodynamics Transport phenomena Water Engineering exergy	
Department	ME Department Process & Energy	

WB2543 Toets 1	*PE&T tentamen*	*3*
Responsible Instructor	Prof.dr.ir. J.T. Padding	
Instructor	Prof.dr.ir. T.J.H. Vlugt	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	Dutch	
Department	ME Department Process & Energy	

WB2543 Toets 2	*Groepswerk + practicum*	*3*
Responsible Instructor	Prof.dr.ir. J.T. Padding	
Instructor	L. Botto	
Co-responsible for assignments	Dr.ir. T.M.J. Nijssen	
Education Period	2	
Start Education	2	
Exam Period	Different, to be announced	
Course Language	Dutch	
Department	ME Department Process & Energy	

WB2633	Advanced Engineering Design Project	6
Responsible Instructor	Dr.ir. A. van Beek	
Instructor	Dr.ir. R.A.J. van Ostayen	
Education Period	1	
Start Education	1	
Course Language	Dutch	
Course Contents	wb2633	
Study Goals	see section descriptions	
Education Method	see section descriptions	
Assessment	see section descriptions	
Department	ME Department Precision & Microsystems Engineering	

WB2633 T1	AED Exam	2
Responsible Instructor	Dr.ir. A. van Beek	
Instructor	Dr.ir. R.A.J. van Ostayen	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	Dutch	
Books	Adv Eng Design + Solutions manual (pdf)	

WB2633 T2	FEM-practical	1
Responsible Instructor	Dr.ir. R.A.J. van Ostayen	
Education Period	1	
Start Education	1	
Exam Period	1	
Course Language	Dutch	

WB2633 T3	AED Project	3
Responsible Instructor	Dr.ir. A. van Beek	
Instructor	Dr.ir. R.A.J. van Ostayen	
Education Period	1	
Start Education	1	
Exam Period	1	
Course Language	Dutch	

Year	**2024/2025**
Organization	**Mechanical Engineering**
Education	**Bachelor Mechanical Engineering**

3e jaar BSc WB 2024

3e jaar WB minors 2024

Year	**2024/2025**
Organization	**Mechanical Engineering**
Education	**Bachelor Mechanical Engineering**

Werktuigbouwkunde blok

WB3135	**Integrated Mechanical Systems**	**6**
Responsible Instructor	Prof.dr.ir. P. Breedveld	
Instructor	Dr.ir. J.F.L. Goosen	
Instructor	Prof. Dr.ir. D.L. Schott	
Instructor	B. van Vliet	
Instructor	Prof.dr.ir. S.A. Klein	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	Dutch	
Department	ME Department Biomechanical Engineering	

WB3135 Toets 1	***IMS - groepsopdracht***	***3***
Responsible Instructor	Prof.dr.ir. P. Breedveld	
Instructor	Dr.ir. J.F.L. Goosen	
Instructor	Prof. Dr.ir. D.L. Schott	
Instructor	B. van Vliet	

WB3135 Toets 2	***IMS - tentamen***	***3***
Responsible Instructor	Prof.dr.ir. P. Breedveld	
Instructor	Dr.ir. J.F.L. Goosen	
Instructor	Prof.dr.ir. S.A. Klein	

WB3240-24	**Systems and Control Engineering**	**6**
Responsible Instructor	Prof.dr.ir. J.W. van Wingerden	
Instructor	Ir. S.C. Bregman	
Instructor	Dr.ir. S.P. Mulders	
Education Period	3	
Start Education	3	
Course Language	Dutch	
Department	ME Department Delft Center for Systems and Control	

WB3240-24 T1	***Systems and Control Engineering***	***2***
Responsible Instructor	Prof.dr.ir. J.W. van Wingerden	
Instructor	Ir. S.C. Bregman	
Instructor	Dr.ir. S.P. Mulders	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	Dutch	

WB3240-24 T2	***Systems and Control Engineering***	***4***
Responsible Instructor	Prof.dr.ir. J.W. van Wingerden	
Instructor	Ir. S.C. Bregman	
Instructor	Dr.ir. S.P. Mulders	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	Dutch	

Year	**2024/2025**
Organization	**Mechanical Engineering**
Education	**Bachelor Mechanical Engineering**

Eindproject blok

WB3073	The engineer in society	4
Responsible Instructor	M. Sand	
Responsible Instructor	O.E. Popa	
Contact Hours / Week x/x/x/x	0/x/0/x	
Education Period	2 4	
Start Education	2 4	
Exam Period	2 4 5	
Course Language	Dutch	
Course Contents	In this course you will explore the ethical and social aspects and problems related to technology and to your future work as professional or manager in the design, development, management or control of technology. You will be introduced to a range of relevant aspects and concepts, including the distinction between descriptive and normative reasoning, philosophical and professional ethics, ethics of robotics, responsibility within organisations, responsible conduct of companies and the role of law. You will analyse ethical, legal and social problems of technology in relation to case studies, and you will explore possibilities for resolving, diminishing or preventing these problems. You will write a text on ethical, social or legal aspect of technology.	
Study Goals	After having completed the course you: can better recognise and analyse ethical and social aspects and problems inherent in technology and in the work of professionals and managers active in the design, development, management and control of technology. have insight into how these ethical and social aspects and problems are related to legal, political and organisational backgrounds. are able to explore and assess possibilities for solving or diminishing existing and emerging ethical and social problems that attach to technology and the work of professionals and managers. can embed ethical and social reasoning into a technical project are better prepared to perform your future work as a professional or manager in the design, development, production and control of technology in an ethical and socially responsible way. can write a short academic essay of critical analysis of an ethical or societal issue related to technology	
Education Method	A series of 7 lectures and the writing of an essay on ethical, social or legal aspects of technology to be written under the supervision of instructors from the section Ethics of Technology at TBM. The essay will be related to and included into the bachelor end project.	
Literature and Study Materials	Articles and documents posted on Brightspace; lectures (slides)	
Assessment	A digital exam (50%) and a written essay (50%) on the ethical, social or legal aspect of technology to be written under the supervision of a lecturer from the Section Ethics/Philosophy to Technology at TBM; the essay will be part of the bachelor end project. Exam and essay can be done in Q2 or Q4. In order to pass the course both the digital exam and the essay should be graded with at least 5 (and the average should be sufficient).	
Enrolment / Application	See Dutch version	

WB3073 Toets 1	*I&S Exam*	*2*

WB3073 Toets 2	*Group Essay*	*2*

WB3BEP-16	Bachelor End Project	14
Responsible Instructor	Dr. R. Delfos	
Instructor	Ir. M.B. Duinkerken	
Instructor	J.W. Haverkort	
Instructor	Dr.ir. M.B. Mendel	
Instructor	Dr. G.J. Verbiest	
Instructor	Dr. P. Dey	
Instructor	B. Fereidoonnezhad	
Instructor	Dr. C. Pek	
Education Period	1 2 3 4	
Start Education	1 3	
Exam Period	none	
Course Language	Dutch	
Education Method	See: http://www.wbmt2.tudelft.nl/bep-wb/BEP_def_EN.pdf There is also a Dutch version; http://www.wbmt2.tudelft.nl/bep-wb/BEP_def_NL.pdf	
Department	ME Department Biomechanical Engineering ME Department Delft Center for Systems and Control ME Department Maritime & Transport Technology ME Department Materials Science & Engineering ME Department Precision & Microsystems Engineering ME Department Process & Energy ME Department Cognitive Robotics	

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Department	Team Kevin Rossi

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